Aakash
Medical | IIT-JEE | Foundations
Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, ***Ph.***+91-1244168300

MM : 720 **Fortnightly Test for NEET-2027 (XII Passed) Ph-2 - FT03B** **Time : 180 Min.**

Topics Covered:
Physics : Work, Energy & Power, System of Particles & Rotational Motion
Chemistry : Chemical Bonding and Molecular Structure
Botany : Biological Classification
Zoology : Breathing & Exchange of Gases-II: (Upto disorders), Body Fluids & Circulation

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

1. A ball collides with a smooth horizontal surface in *xz*-plane with a velocity $\vec{v} = 2\hat{i} - 3\hat{j}$. If the coefficient of restitution is $\frac{1}{3}$, then the velocity of the ball just after the impact will be

(1) $-\frac{2}{3}\hat{i} - 3\hat{j}$ (2) $\frac{1}{3}\left(2\hat{i} + 3\hat{j}\right)$

(3) $2\hat{i} + \hat{j}$ (4) $\frac{2}{3}\hat{i} - \hat{j}$

2. The potential energy of a long spring when stretched by 2 cm, is *U*. If the spring is stretched by 8 cm, increase in potential energy in the spring will be

(1) 16 *U* (2) 4 *U*
(3) 15 *U* (4) 12 *U*

3. A : Centripetal force cannot do any work on the object moving in a circle.
R : Centripetal force and instantaneous velocity always remain parallel to each other in circular motion.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

4. An engine pumps 800 kg of water through height of 10 m in 80 s. The power of the engine if its efficiency is 75%, is approx

(1) 1.3 kW (2) 1.5 kW
(3) 2.1 kW (4) 1.8 kW

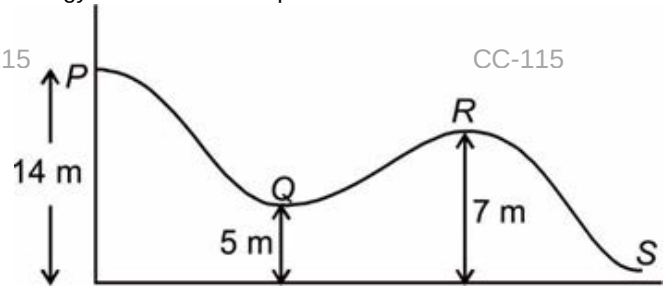
5. Kinetic energy of a particle moving in straight line varies with time t as $k = 9t^2$. The power of the particle

(1) Is constant
(2) Increasing with time (*t*)
(3) Decreasing with time (*t*)
(4) First increases and then decreases

6. A small block of mass m connected to a massless and unstretchable string goes in a vertical circle of radius R under gravity. The other end of the string is fixed at the centre of the circle. If the velocity at the top of the circular path is $3\sqrt{gR}$, then the ratio of the kinetic energy of the block at the bottom to that at the top of the circle is

(1) $\frac{12}{11}$ (2) $\frac{13}{9}$
(3) $\frac{14}{9}$ (4) $\frac{15}{8}$

7. Figure shows the vertical section of frictionless surface. A block of mass 2 kg is released from the position *P*, its kinetic energy as it reaches the position *R* is



(1) 108 J (2) 40 J
(3) 140 J (4) 280 J

8. A body falls on a surface of coefficient of restitution 0.6 from a height of 1 m. Then the body rebounds to a height of
(1) 0.6 m (2) 0.4 m
(3) 1 m (4) 0.36 m

9. Two masses of 0.25 kg each moves toward each other with speed 3 ms^{-1} and 1 ms^{-1} collide and stick together. Find the final velocity.
(1) 0.5 ms^{-1} (2) 2 ms^{-1}
(3) 1 ms^{-1} (4) 0.25 ms^{-1}

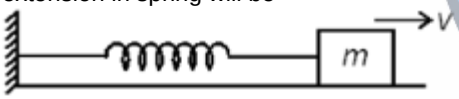
10. If the momentum of a body is increased by 20% then its kinetic energy increased by
(1) 20% (2) 40%
(3) 44% (4) 48%

11. A 2 kg block moving with velocity 6 m/s collides with stationary 4 kg block. The lighter block comes to rest after collision. The loss in kinetic energy of system is
(1) 9 J (2) 18 J
(3) 20 J (4) 25 J

12. **Statement-A :** Work done by frictional force is always negative.
Statement-B : A body at rest can have mechanical energy.
(1) Both statements are correct
(2) Both statements are incorrect
(3) Statement A is correct and B is incorrect
(4) Statement A is incorrect and B is correct

13. A force $F = 10 + 2x$ acts on a particle moving in straight line on x-axis. The work done by this force during a displacement from $x = 0$ to $x = 3$ m is
(1) 13 J (2) 29 J
(3) 39 J (4) 43 J

14. A block of mass m is connected to one end of a spring of spring constant K, whose other end is fixed to a wall, as shown in the figure. If initially the spring is unstretched and block is given an initial velocity v, then the maximum extension in spring will be

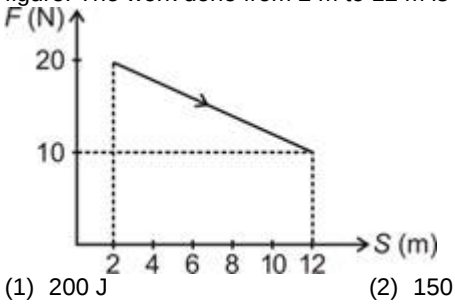


(1) $\sqrt{Km}\,v$ (2) $\sqrt{\frac{K}{m}}\,v$
(3) $\sqrt{\frac{m}{K}}\,v$ (4) Kmv

15. When a conservative force does positive work on a body then
(1) Its kinetic energy must increase
(2) Its potential energy must decrease
(3) Its potential energy must increase
(4) Its total energy must decrease

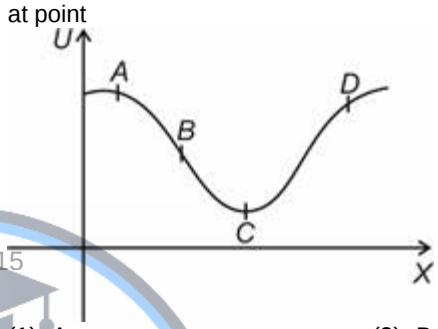
16. Two springs have force constants K_1 and K_2 such that $K_1 > K_2$, if the springs are stretched by same force, work done for 1st spring is W_1 and work done for 2nd spring is W_2 .*Then*
(1) $W_1 = W_2$ (2) $W_1 > W_2$
(3) $W_1 < W_2$ (4) $W_1 = 0, W_1 \neq 0$

17. The force-displacement graph of a body is as shown in figure. The work done from 2 m to 12 m is



(1) 200 J (2) 150 J
(3) 100 J (4) 175 J

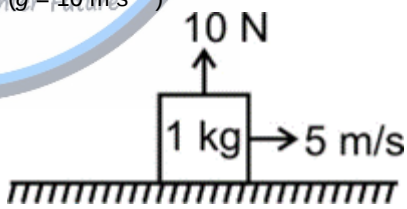
18. Potential energy (U) - position (X) graph of a body is as shown in the figure. The particle can attain stable equilibrium at point



(1) A (2) B
(3) C (4) D

19. The potential energy of a conservative system is given by relation $U = ax^2 - bx$, where x represents position of the particle and a and b are constant. What is the force acting on the system?
(1) $2ax - b$ (2) $ax + b$
(3) $-2ax + b$ (4) $ax^3 + bx^2$

20. A force of 10 N is applied on a moving block placed on smooth surface as shown in the figure. The change in kinetic energy as the block displaces from $x = 4$ m to $x = 5$ m will be ($g = 10\ m\ s^{-2}$)

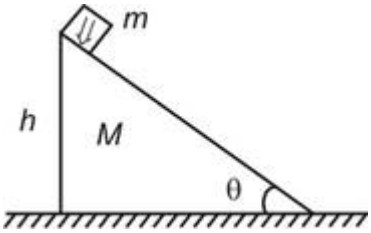


(1) 5 J (2) 10 J
(3) 2.5 J (4) Zero

21. Two identical particles moving with same speed (v) in different directions collide inelastically and after perfectly inelastic collision speed of the particles becomes half the initial speed v/2 . Find the angle between the two before collision.
(1) 60° (2) 30°
(3) 120° (4) 90°

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22. If system is released from rest and all surfaces are smooth then displacement of wedge when small block reach at the bottom will be



(1) $\frac{mh}{m+M}$ (2) $\frac{mh\tan\theta}{m+M}$
(3) $\frac{mh\cos\theta}{m+M}$ (4) $\frac{mh\cot\theta}{m+M}$

23. A thin circular ring of mass *M* and radius *r* is rotating about its axis with an angular speed ω. Two particles having mass *m* each are now gently attached at diametrically opposite points. The angular speed of the ring will become
(1) $\frac{\omega M}{M+m}$ (2) $\frac{\omega M}{M+2m}$
(3) $\frac{\omega(M+3m)}{m}$ (4) $\frac{\omega(M+2m)}{M}$

24. Match the physical quantity given in **List-I** with their respective formula in **List-II** and choose the correct option.

	List-I (Quantity)		**List-II (Corresponding formula)**
A.	Torque τ	I.	$M\vec{a}_{cm}$
B.	Angular momentum	II.	I α
C.	$\sum\vec{F}_{ext}$	III.	$Mvr_{\perp}$
D.	Instantaneous rotational power	IV.	$\vec{\tau}\cdot\vec{\omega}$

Chose the correct answer from the options given below:
(1) A-III, B-II, C-I, D-IV (2) A-III, B-II, C-IV, D-I
(3) A-II, B-III, C-I, D-IV (4) A-II, B-III, C-IV, D-I

25. Two rings have moment of inertia about their centre in the ratio of 2 : 1. Their diameters are in the ratio of 2 : 1. The ratio of their masses will be
(1) 2 : 1 (2) 1 : 2
(3) 1 : 4 (4) 1 : 1

26. The radius of gyration of a uniform solid sphere about a tangent is
(1) $R\sqrt{\frac{2}{5}}$ (2) $R\sqrt{\frac{2}{3}}$
(3) $R\sqrt{\frac{5}{3}}$ (4) $R\sqrt{\frac{7}{5}}$

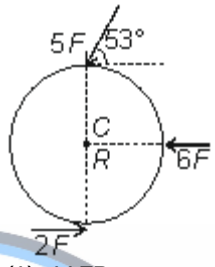
27. If the rotational kinetic energy of a body is increased by 800%, then the percentage increase in its angular momentum is
(1) 100% (2) 200%
(3) 300% (4) 283%

28. If the net external force acting on the system of particles is zero, then which of the following may vary?
(1) Momentum of the system
(2) Kinetic energy of the system
(3) Acceleration of centre of mass
(4) All of these

29. 3 identical thin rods each of length 'L' and mass M are joined together to form a letter 'H'. The moment of inertia of a system about one of the vertical sides of 'H' is
(1) $\frac{ML^2}{4}$ (2) $\frac{ML^2}{3}$
(3) $\frac{2ML^2}{3}$ (4) $\frac{4ML^2}{3}$

30. A particle of mass *m* moving with speed v_0 along a line $y = 2x + 4$, the angular momentum of particle about origin is
(1) Continuously increasing (2) Continuously decreasing
(3) Remains constant (4) Zero

31. Three forces are acting on a wheel (all forces are in the plane of wheel) as shown. Net torque about point *C*(centre) of the disc is



(1) 11*FR* (2) 5*FR*
(3) 6*FR* (4) *FR*

32. If a boy standing on a stationary boat, moves on the boat to his right, then the
(1) Boat moves to his right
(2) Boat remains stationary
(3) Centre of mass of the system (boat + boy) moves to the left
(4) Centre of mass of the system (boat + boy) remains stationary

33. If axis of rotation is given by a line in *xy* plane as $y = -x$. Then torque of a force *F* = 10 N, about the given line acting parallel to this line at a separation of 2 m from it, is
(1) 0 (2) 20 N m
(3) 10 N m (4) 5 N m

34. The value of $\frac{K^2}{R^2}$ for a solid sphere with radius *R* and radius of gyration *K* about an axis along its diameter is
(1) 1 (2) $\frac{2}{5}$
(3) $\frac{1}{2}$ (4) $\frac{2}{3}$

35. A rigid body from rest starts rotating about fixed axis with a constant angular acceleration. During 2nd second the body turns by an angle 5θ. In 5th second, the body will turn by an angle
(1) 15θ (2) 16θ
(3) 7θ (4) 28θ

36. The moment of the force, $\vec{F}=\hat{i}-2\hat{j}+\hat{k}$ at (1, 2, –1), about the origin, is given by
(1) $-2\hat{i}+4\hat{k}+0\hat{j}$ (2) $0\hat{i}-2\hat{j}-4\hat{k}$
(3) $\hat{i}-4\hat{j}-\hat{k}$ (4) $\hat{i}+2\hat{j}-\hat{k}$

37. Two discs *A* and *B* have same mass and same thickness. If d_1 and d_2 are the densities of the materials of the discs *A* and *B* respectively, then the ratio of the moment of inertia of the discs *A* and *B* about their geometrical axis is

(1) $\frac{d_1^2}{d_2^2}$

(2) $\frac{d_2^2}{d_1^2}$

(3) $\frac{d_1}{d_2}$

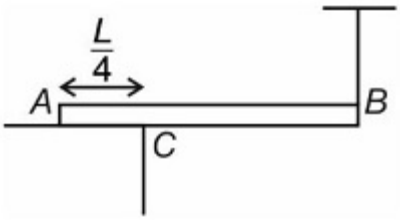
(4) $\frac{d_2}{d_1}$

38. A man of mass M stands at the edge of a circular platform of radius R and moment of inertia I. The platform is at rest initially. The platform begins to rotate when the man begins to move along the tangent with velocity v w.r.t. ground. The angular velocity of the platform is

(1) $\frac{MvR}{I}$ (2) $\frac{MvR}{I-MR^2}$

(3) $\frac{MvR}{I+MR^2}$ (4) $\frac{MvR}{I+2MR^2}$

39. A uniform rod of mass *m* and length *L* is kept on horizontal table with $\frac{L}{4}$ length on the table as shown in figure. As the string attached to the end *B* is cut, the angular acceleration of the rod about point *C* will be



(1) $\frac{12g}{7L}$ (2) $\frac{9g}{7L}$

(3) $\frac{4g}{5L}$ (4) $\frac{3g}{7L}$

40. **Assertion(A):** If moment of inertia of a non-uniform thin circular ring is same about two different axes parallel to each other and lying in the plane of ring, then both the axes can be at same distance from centre of mass of ring.
Reason(R): From parallel axes theorem, $I_1 = I_2$ is possible only if

$I_{cm} + ml_1^2 = I_{cm} + ml_2^2$

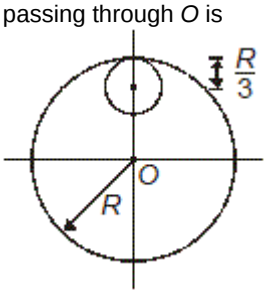
In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

41. A metre stick is balanced on a knife edge at its centre when two coins each of mass 5 g are put one on the other at the 12 cm mark, the stick is found to be balanced at 45 cm. The mass of the metre stick is

(1) 56 g (2) 66 g
(3) 76 g (4) 36 g

42. From a circular disc of radius *R* and mass *M*, a small disc of radius $\frac{R}{3}$ is removed. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through *O* is



(1) $\frac{MR^2}{18}$ (2) $\frac{4}{9}MR^2$

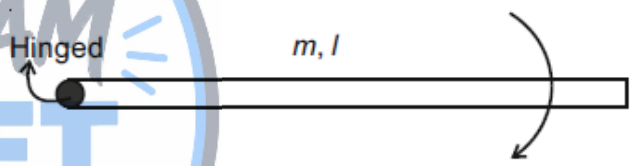
(3) $\frac{5MR^2}{18}$ (4) $\frac{5}{9}MR^2$

43. Two semicircular rings of linear mass densities λ and 2 λ and of radius '*R*' each are joined to form a complete ring. The distance of the center of the mass of complete ring from its geometrical centre is

(1) $\frac{3R}{8\pi}$ (2) $\frac{3R}{4\pi}$

(3) $\frac{2R}{3\pi}$ (4) $\frac{4R}{4\pi}$

44. A uniform rod of mass m and length *l* is fixed at one end released from horizontal position as shown in the figure. The angular velocity of the rod when the rod makes an angle 60° with vertical is



(1) $\sqrt{\frac{3g}{2l}}$ (2) $\sqrt{\frac{4g}{3l}}$

(3) $\sqrt{\frac{2g}{l}}$ (4) $\sqrt{\frac{3g}{l}}$

45. Read the following statements carefully and choose the correct option.
Statement (A): Kinetic energy of a system just before elastic collision is equal to kinetic energy of the system just after the collision.
Statement (B): Linear momentum of a system just before elastic collision is equal to linear momentum of the system just after the collision.
Statement (C): Linear momentum of system just before inelastic collision is equal to linear momentum of system just after the collision.

(1) All statements (A), (B) and (C) are correct
(2) Only statements (A) and (B) are correct
(3) Only statement (B) is correct
(4) Only statement (A) is correct

CHEMISTRY

46. The incorrect statement regarding octet rule is
(1) It does not account for the shape of molecules
(2) It does not explain the relative stability of the molecules
(3) It is based upon the chemical inertness of noble gases
(4) It is mainly applicable to the elements beyond the third period of the periodic table

47. An example of odd-electron molecule is
(1) F_2 (2) BCl_3
(3) NO_2 (4) O_3

48. Formal charge on the central oxygen atom in O_3 molecule is
(1) Zero (2) –1
(3) +1 (4) –2

49. Consider the following species:
(a) PF_5
(b) SF_6
(c) H_2SO_4
(d) NO_2
(e) CCl_4
The expanded octet is observed in
(1) (c), (d) and (e) only (2) (a), (b) and (c) only
(3) (b), (c) and (d) only (4) (a), (b) and (e) only

50. Which among the following has highest melting point?
(1) NaF (2) NaCl
(3) NaBr (4) NaI

51. Which among the following has highest lattice enthalpy?
(1) LiCl (2) NaCl
(3) LiI (4) NaI

52. Pure covalent molecule is
(1) HCl (2) NH_3
(3) CO_2 (4) N_2

53. Maximum C – O bond length is in
(1) CO (2) CO_3^{2-}
(3) CO_2 (4) C_3O_2

54. If the Dipole moment of AB molecule is 2.5 D and A–B bond length is 1.5 Å then the percentage ionic character of AB molecule will be
(1) 24.2% (2) 52.1%
(3) 18.4% (4) 34.7%

55. Cl–O bond order in ClO_4^- ion is
(1) 1.25 (2) 1.50
(3) 1.66 (4) 1.75

56. The correct order of bond length of C – C, C – N, C – H is
(1) C – H < C – N < C – C (2) C – C < C – N < C – H
(3) C – C < C – H < C – N (4) C – N < C – H < C – C

57. Consider the following statements:
(i) NaCl is more covalent than $MgCl_2$
(ii) CsI is more ionic as compared to NaI
(iii) Al^{3+} has more polarising power than Mg^{2+}
The **incorrect** statement(s) is/are
(1) (i) only (2) (ii) only
(3) (ii) & (iii) only (4) (i) & (iii) only

58. Total number of sigma bonds in acetylene molecule is
(1) 1 (2) 2
(3) 3 (4) 5

59. Select the incorrect statement(s) among the following
(I) ClF_3 is sp^3d hybridised
(II) The equatorial bonds are longer than axial bonds in PCl_5
(III) All the bonds are equivalent in SF_6 molecule
(1) I only (2) III only
(3) II only (4) II and III only

60. Which among the following molecule is not planar?
(1) $BeCl_2$ (2) SO_3
(3) XeF_4 (4) SF_4

61. Select the halide of highest ionic character
(1) NaF (2) NaCl
(3) NaBr (4) NaI

62. Decreasing order of O–O bond length in the given species is
(1) $O_2^+ > O_2^- > O_2$ (2) $O_2 > O_2^+ > O_2^-$
(3) $O_2^+ > O_2 > O_2^-$ (4) $O_2^- > O_2 > O_2^+$

63. Match Column I with Column II.

	Column I Species		**Column II Shape**
a.	SO_3	(i)	Octahedral
b.	BrF_3	(ii)	Trigonal planar
c.	XeO_3	(iii)	Bent T-shape
d.	SF_6	(iv)	Trigonal pyramidal

Choose the correct answer from the options given below
(1) a(ii), b(iii), c(i), d(iv) (2) a(i), b(iv), c(iii), d(ii)
(3) a(ii), b(iii), c(iv), d(i) (4) a(iv), b(iii), c(ii), d(i)

64. The correct order of H–N–H bond angle in NH_2^-, NH_3 and NH_4^+ is
(1) $NH_2^- > NH_3 > NH_4^+$ (2) $NH_4^+ > NH_3 > NH_2^-$
(3) $NH_3 > NH_4^+ > NH_2^-$ (4) $NH_4^+ > NH_2^- > NH_3$

65. Match List-I with List-II

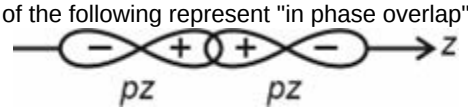
	List-I (Species)		**List-II (Bond angle)**
(a)	BF_3	(i)	Contains only 180° and 90° bond angles
(b)	XeF_4	(ii)	Contains only 120° bond angle
(c)	PCl_5	(iii)	Contains 90°, 180° and 120° bond angles
(d)	CO_2	(iv)	Contains only 180° bond angle

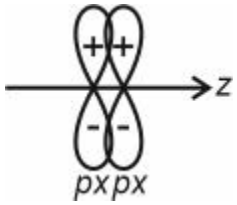
The correct match is
(1) a(ii), b(iii), c(i), d(iv) (2) a(i), b(iii), c(ii), d(iv)
(3) a(ii), b(i), c(iii), d(iv) (4) a(i), b(ii), c(iv), d(iii)

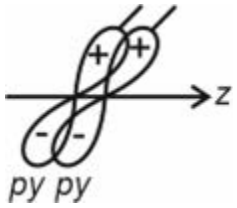
66. Which among the following species has $p\pi - p\pi$ overlapping?
(1) H_2 (2) CO_2
(3) BeH_2 (4) H_2O

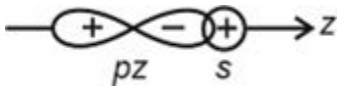
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67. All of the following represent "in phase overlap", **except**

(1) 

(2) 

(3) 

(4) 

68. Which of the following species contains equal number of σ and π bonds?

(i) XeO_3

(ii) SO_3

(iii) SO_2

(iv) $CH_2(CN)_2$

Choose the **correct** option

(1) (i), (iii) and (iv) only (2) (ii) and (iv) only

(3) (i), (ii) and (iii) only (4) (iii) only

69. Given below are two statements: one is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion (A): Electrons present in antibonding molecular orbital destabilize the molecule.

Reason (R): Mutual repulsion of the electrons in antibonding molecular orbital is more than the attraction between the electrons and the nuclei which causes a net increase in energy.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true and R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

70. The change in the hybridisation of B as result of the following reaction is

$BF_3 + NH_3 \rightarrow BF_3 \leftarrow NH_3$

(1) sp^2 to sp^3 (2) sp^2 to sp

(3) sp to sp^3 (4) sp^3 to sp^2

71. Which among the following features of hybridisation is correct?

(1) The pure atomic orbitals are more effective in forming stable bonds than the hybrid orbitals

(2) The number of hybrid orbitals is not always equal to the number of the atomic orbitals that get hybridised

(3) The hybridised orbitals are always equivalent in energy and shape

(4) These hybrid orbital are non-directional in nature

72. The state of hybridisation of C_1, C_3, C_5 in hydrocarbon CH ≡ C – CH = CH – CH_2 – CH_3 is respectively

(1) sp^3, sp, sp^2 (2) sp, sp^3, sp^2

(3) sp, sp^2, sp^3 (4) sp, sp, sp^3

73. Which of the following orbital overlap results into formation of C-C bond with highest bond energy?

(1) sp^3 - sp^3 (2) sp^2 - sp^2

(3) sp^3 - sp^2 (4) sp - sp

74. The percentage of p-character in sp^2 hybridisation is approximately

(1) 25% (2) 33%

(3) 66% (4) 75%

75. Which species has the maximum number of lone pair of electrons on the central atom?

(1) ClO_3^- (2) XeF_4

(3) SF_4 (4) I_3^-

76. The electronic configuration of O_2 molecule is

(1) $(\sigma 1s)^2(\sigma^* 1s)^2\ (\sigma 2s)^2\ (\sigma^* 2s)^2\ \left(\pi 2p_x^2 \equiv \pi 2p_y^2\right)$

$\left(\pi^* 2p_x^2 \equiv \pi^* 2p_y^2\right)$

(2) $(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p_z)^2\ \left(\pi 2p_x^2 \equiv \pi 2p_y^2\right)$

$\left(\pi^* 2p_x^1 \equiv \pi^* 2p_y^1\right)$

(3) $(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p_z)^2(\ \pi 2p_x^1 \equiv \pi 2p_y^1)$

$\left(\pi^* 2p_x^2 \equiv \pi^* 2p_y^2\right)$

(4) $(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2\ \left(\pi 2p_x^2 \equiv \pi 2p_y^2\right)$

$\left(\pi^* 2p_x^1 \equiv \pi^* 2p_y^1\right)\ \sigma^* 2\ p_z^2$

77. The species which does not exist is

(1) He_2 (2) B_2

(3) O_2^- (4) N_2^+

78. In which of the following species the highest occupied molecular orbital is σ*?

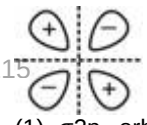
(1) N_2 (2) C_2

(3) O_2^- (4) He_2^+

79. Compare the stability of the following species : H_2, H_2^+ and H_2^-

(1) $H_2^- < H_2^+ < H_2$ (2) $H_2 < H_2^+ < H_2^-$

(3) $H_2^+ < H_2^- < H_2$ (4) $H_2^+ < H_2 < H_2^-$

80.  This molecular orbital is representing:

(1) $\sigma 2p_z$ orbital (2) $\pi^* 2p_x$ orbital

(3) $\sigma^* 2p_z$ orbital (4) $\pi 2p_x$ orbital

81. During change of N_2 to N_2^- ion, the electron adds to which of the following molecular orbitals?

(1) σ^* orbital (2) π^* orbital

(3) σ orbital (4) π orbital

82. In which of the given molecules only π bonds are present?

(1) O_2 (2) C_2

(3) N_2 (4) H_2

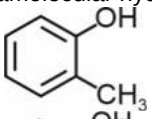
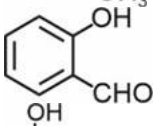
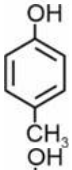
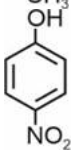
83. Which of the following is diamagnetic in nature?
(1) O_2 (2) O_2^+
(3) O_2^- (4) O_2^{2-}

84. Given below are the two statements:
Statement I: Li_2 and F_2 have same bond order.
Statement II: Bond length of F_2 is longer than Li_2.
In the light of above statements, choose the correct answer from the options given below.
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

85. Considering x axis as the intermolecular axis, which of the following will not form sigma bond?
(1) 2s and 2s (2) $2p_x$ and 2s
(3) $2p_x$ and $2p_x$ (4) $2p_y$ and $2p_y$

86. The compound which involves hydrogen bonding is
(1) PH_3 (2) NH_3
(3) CH_4 (4) H_2S

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87. Intramolecular hydrogen bond is observed in
(1) 
(2) 
(3) 
(4) 

88. Hybridisation of nitrogen in NH_4^+ ion is
(1) *sp* (2) sp^2
(3) sp^3 (4) sp^3d

89. Which of the following pairs of species are isoelectronic and isostructural?
(a) CO_3^{2-}, NO_3^-
(b) SO_3^{2-}, NO_3^-
(c) ClO_3^-, CO_3^{2-}
(d) ClO_3^-, SO_3^{2-}
Choose the correct option
(1) (a) and (b) only (2) (b) and (c) only
(3) (c) and (d) only (4) (a) and (d) only

90. Which of the following is the correct order of dipole moments?
(1) $BF_3 > NH_3 > NF_3$ (2) $NH_3 > NF_3 > BF_3$
(3) $NH_3 > BF_3 > NF_3$ (4) $NF_3 > NH_3 > BF_3$

BOTANY

91. Read the following statements (A – D).
(A) Cyanobacteria are photosynthetic prokaryotes, being the most primitive organism to have oxygenic photosynthesis.
(B) Bacterial structure is very simple but they are very complex in their behaviour.
(C) Smallest wall-less monerans are mostly autotrophic.
(D) Archaebacteria differ from other bacteria in having a different cell wall structure which is responsible for their survival in extreme conditions.
Select the option for the **correct** set of statements.
(1) Only (A), (B) and (C) (2) Only (B) and (D)
(3) Only (C) and (D) (4) Only (A), (B) and (D)

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92. Observe the following diagram.



Select the **correct** statement for this organism.
(1) This organism can oxidise various inorganic substances and use the released energy for their ATP production.
(2) It is helpful in production of antibiotics such as penicillin.
(3) It reproduces sexually by a process involving cell fusion and zygote formation.
(4) This organism can fix both atmospheric nitrogen and carbon dioxide.

93. Read the following assertion (A) and reason (R) statements and select the **correct** option.
Assertion(A): In the five kingdom classification of Whittaker, there is no mention of acellular organisms like viruses.
Reason(R): Viruses are not considered truly, 'living', if we understand living as those organisms that have a cell structure.
(1) Both (A) and (R) are false
(2) Both (A) and (R) are true and (R) is the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

94. The causative agent of potato spindle tuber disease is
(1) Virus (2) Prion
(3) Viroid (4) Lichen

95. Which one is **incorrect** w.r.t. lichens?
(1) Lichens are symbiotic association between algae and fungi
(2) Algal component is known as phycobiont
(3) Algae prepares food and fungi in return provides shelter, absorbs mineral nutrients and water for its partner
(4) Lichens are very good pollution indicators as they grow well in polluted areas

96. Match **List-I** with **List-II**.

	List-I		List-II
A.	Fusion of protoplasms between gametes	I.	Meiosis
B.	Fusion of two nuclei	II.	Plasmogamy
C.	Generation of haploid spores	III.	Karyogamy

Choose the **correct** answer from the options given below :
(1) A-II, B-III, C-I (2) A-II, B-I, C-III
(3) A-III, B-II, C-I (4) A-I, B-III, C-II

97. Read the following statements and mark the **correct** option.
Statement A : Bacteria as a group show the most extensive metabolic diversity.
Statement B : Bacteria are the sole members of the kingdom Monera.
(1) Only A is incorrect
(2) Only B is incorrect
(3) Both A and B are correct
(4) Both A and B are incorrect

98. The vast majority of bacteria are
(1) Chemosynthetic autotrophs
(2) Heterotrophs
(3) Photoautotrophs
(4) Capable of synthesizing their own food

99. Identify the organisms on the basis of given information and select the **correct** option.
• They are the chief producers in the oceans.
• Their walls are embedded with silica.
(1) Euglenoids (2) *Mycoplasma*
(3) Methanogens (4) Diatoms

100. The basidiocarps are edible in
(1) Truffles (2) Morels
(3) *Agaricus* (4) Yeast

101. Which of the following is **not** the sexual spores in fungi?
(1) Oospores (2) Ascospores
(3) Conidia (4) Basidiospores

102. Select the **incorrect** statement w.r.t. dianoflagellates
(1) May appear yellow, green, brown, blue or red in colour
(2) Some member releases toxin that may even kill other marine animals
(3) Have two flagella, both lie transversely
(4) They are mostly marine organism

103. Slime moulds
(1) Are multicellular organisms
(2) Form plasmodium under unfavourable conditions
(3) Are saprophytic fungi
(4) Produce spores which possess true walls

104. Identify the **wrongly** matched pair.
(1) Rod shaped bacteria Bacilli
(2) Comma shaped bacteria Vibrio
(3) Spherical bacteria Desmids
(4) Spiral bacteria Spirilla
(1) (1) (2) (2)
(3) (3) (4) (4)

105. Select the **incorrect** one for prions.
(1) Similar in size to viruses
(2) Consist of wrongly folded proteins
(3) Cause certain infectious neurological diseases
(4) Have RNA as genetic material

106. Read the statements I and II and select the correct option.
Statement I: All the unicellular organisms are placed in kingdom Protista.
Statement II: Kingdom Protista has brought together unicellular photosynthetic organisms like *Chlamydomonas*, *Chlorella*, *Spirogyra* with *Paramoecium* and *Amoeba*.
In the light of above statements, choose the **most appropriate** answer from the options given below:
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

107. Match the following columns and select the **correct** option.

	Column I		Column II
a.	Viroids	(i)	Symbiotic association with roots of higher plants
b.	Mycorrhiza	(ii)	Contain either DNA or RNA
c.	Viruses	(iii)	Infectious RNA particles without protein coat
d.	Prions	(iv)	Cause mad cow disease in cattle

(1) a(ii), b(iii), c(i), d(iv) (2) a(iii), b(i), c(ii), d(iv)
(3) a(ii), b(iv), c(iii), d(i) (4) a(iv), b(i), c(iii), d(ii)

108. Euglenoids have
(1) Two flagella of same size
(2) Lipid rich layer called pellicle
(3) Pigments identical to those present in higher plants
(4) Heterotrophic mode of nutrition only in the presence of sunlight

109. Tetanus is caused by a
(1) Protozoan (2) Fungus
(3) Bacterium (4) Virus

110. Select the **incorrect** statement w.r.t. TMV.
(1) Genetic material is ssDNA
(2) D.J. Ivanowsky's work was associated with it
(3) It can pass through bacterial filter
(4) Rod shaped virus with helical symmetry

111. Select the **odd** one w.r.t. characteristics of the members of kingdom Fungi.
(1) They have prominent nucleus
(2) They are multicellular with loose tissue body organisation
(3) All of them have mycelia that consist of septate hyphae
(4) They have heterotrophic mode of nutrition

112. Outermost layer is **not** cell wall in
A) *Gonyaulax*
B) *Euglena*
C) *Paramoecium*
D) *Amoeba*
The **correct** ones are
(1) A only (2) A, B & C
(3) B, C & D (4) B & C only

113. *Paramoecium* and *Plasmodium*
(1) Both belong to the same kingdom as that of *Penicillium*
(2) Both have cilia as locomotory organs
(3) Both are endoparasites
(4) Both are unicellular protozoans

114. The parasitic fungus on mustard is
(1) *Puccinia* (2) *Ustilago*
(3) *Rhizopus* (4) *Albugo*

115. Read the following features and select the option for which these features are **correct**.
a. Obligate anaerobes
b. Chemoautotrophic mode of nutrition
c. Found in marshy habitats
(1) Blue green algae (2) Slime moulds
(3) Methanogens (4) Dinoflagellates

116. Deuteromycetes are also called 'Fungi imperfecti' because
(1) They do not recycle minerals perfectly
(2) They do not reproduce sexually
(3) They lack sex organs but reproduce sexually
(4) They have non-chitinous cell wall

117. *Trypanosoma* is a
(1) Free-living flagellated protozoan
(2) Parasitic ciliated protozoan
(3) Parasitic flagellated protozoan
(4) Parasitic sporozoan

118. In five kingdom classification, *Chlorella* is placed in kingdom
(1) Protista (2) Monera
(3) Plantae (4) Animalia

119. In *Nostoc*, nitrogen fixation occurs in
(1) All the living cells
(2) Some specialised cells called heterocyst
(3) All the cells except heterocyst
(4) Only those cells which have thin cell wall

120. Aristotle used simple morphological characters to classify plants into
(1) Aquatic and terrestrial plants
(2) Herbs, shrubs and trees
(3) Xerophytic, mesophytic and hydrophytic plants
(4) Flowering and non-flowering plants

121. Earliest classification system included *Chlamydomonas* and *Spirogyra* in
(1) Protista (2) Monera
(3) Plantae (4) Fungi

122. Mycelium as continuous tubes filled with multinucleated cytoplasm is found in
(1) *Saccharomyces* and *Penicillium*
(2) *Rhizopus*, *Mucor* and *Albugo*
(3) *Rhizopus*, *Ustilago* and *Puccinia*
(4) *Alternaria*, *Colletotrichum* and *Albugo*

123. Read the following statements and state true (T) or false (F) and choose the **correct** option.
A. In *Paramoecium*, the coordinated movement of rows of cilia causes the water laden with food to be steered into the gullet.
B. All members of kingdom fungi are filamentous.
(1) **A** – F, **B** – T (2) **A** – T, **B** – F
(3) **A** – F, **B** – F (4) **A** – T, **B** – T

124. Sleeping sickness in human is caused by
(1) *Paramoecium* (2) *Amoeba*
(3) *Entamoeba* (4) *Trypanosoma*

125. The scientist who called the extract of infected tobacco plant as *Contagium vivum fluidum* was
(1) D.J. Ivanowsky (2) Pasteur
(3) W.M. Stanley (4) M.W. Beijerinek

126. Which fungi is used extensively in biochemical and genetic work?
(1) *Agaricus* (2) *Neurospora*
(3) Morels (4) *Trichoderma*

127. Red tide is caused by
(1) Desmids (2) Diatoms
(3) *Gonyaulax* (4) A cyanobacterium

128. Rusts and smuts in wheat are diseases caused by members of
(1) Slime moulds (2) Protozoans
(3) Deuteromycetes (4) Basidiomycetes

129. Glycogen is reserve food in
(1) Fungi (2) Desmids
(3) *Gonyaulax* (4) Mango plant

130. Kuru disease is caused by
(1) Viroids (2) Virusoids
(3) Prions (4) Viruses

131. Loose tissue body organisation is a characteristic feature of
(1) Monera (2) Protista
(3) Fungi (4) Plantae

132. Read the following statements for slime moulds and select the correct option.
(a) They have naked protoplast in vegetative stage
(b) Their spores may survive for many years
(c) In favorable condition they form fruiting bodies
(d) They reproduce only by asexual spores
Select the **correct** option.
(1) Only (a) and (c) are correct
(2) Only (a), (b) and (c) are correct
(3) Only (c) and (d) are incorrect
(4) All (a), (b), (c) and (d) are correct

133. Read the following statements and choose the **correct** option.
Assertion : Kingdom Protista has brought cell wall containing algae along with organisms without cell wall under five kingdom classification.
Reason : Criteria of classification were changed from Linnaeus's classification to Whittaker's classification.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion.
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion.
(3) Assertion is true statement but Reason is false.
(4) Both Assertion and Reason are false statements.

134. *Cuscuta* is
(1) A partially heterotrophic plant
(2) A parasitic plant
(3) An insectivorous plant
(4) A prokaryote

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135. All _____ are obligate parasites
(1) Fungi (2) Animals
(3) Viruses (4) Protists

ZOOLOGY

136. Heart attack is defined as the condition when
(1) Heart stops beating
(2) Heart muscle damaged suddenly due to inadequate blood supply
(3) Heart fails to pump blood effectively
(4) Congestion of lungs is the main symptom

137. Difficulty in breathing causing wheezing due to inflammation of bronchi and bronchioles is characteristic finding of
(1) Asthma (2) Emphysema
(3) Asbestosis (4) Silicosis

138. The normal count of platelets in human blood is
(1) 6000 - 8000 mm^{-3}
(2) 5 to 5.5 millions mm^{-3}
(3) 1,50,000 to 3,50,000 mm^{-3}
(4) 10,000 to 1,00,000 mm^{-3}

139. Incomplete double circulation is present in
(1) Cow (2) Parrot
(3) Horse (4) Snake

140. The first heart sound is associated with
(1) Opening of the tricuspid and bicuspid valves.
(2) Closure of the tricuspid and bicuspid valves.
(3) Opening of the semilunar valves.
(4) Closure of the semilunar valves.

141. Pacemaker of the heart is situated in the
(1) Right upper corner of the left atrium
(2) Lower left corner of the left atrium
(3) Right upper corner of the right atrium
(4) Lower left corner of the right atrium

142. A reduction in the number of which of the following can lead to clotting disorders that lead to excessive loss of blood from the body?
(1) Platelets (2) Erythrocytes
(3) Neutrophils (4) Monocytes

143. Maximum ventricular filling occurs when
(1) Ventricles relax and atria contract
(2) Ventricles contract and atria relax
(3) Both ventricles and atria contract
(4) Both ventricles and atria relax

144. The shape of oxygen dissociation curve is
(1) Circular (2) Parabolic
(3) Sigmoid (4) Pyramidal

145. Body of a person contains 5 L blood. What amount of O_2 can be delivered by 5 L of blood to tissues under normal physiological conditions?
(1) 750 mL (2) 500 mL
(3) 250 mL (4) 150 mL

146. Partial pressure of oxygen in oxygenated blood under normal physiological conditions is
(1) 40 mmHg (2) 95 mmHg
(3) 45 mmHg (4) 159 mmHg

147. During blood coagulation, the conversion of fibrinogen to fibrin is caused by an enzyme 'E'. Select the option that represents the inactive form of 'E'.
(1) Thrombokinase (2) Prothrombin
(3) Thrombin (4) Fibrin

148. In ECG of human heart, QRS complex denotes
(1) Repolarisation of atrium
(2) Depolarisation of atrium
(3) Repolarisation of ventricle
(4) Depolarisation of ventricle

149. Choose the **incorrect** statement.
(1) Plasma is a viscous fluid constituting nearly 55% of the blood
(2) Fibrinogen is involved in clotting of the blood
(3) Plasma with active clotting factors is called serum
(4) Formed elements constitute nearly 45 per cent of blood

150. pO_2 in pulmonary artery is
(1) Equal to pO_2 in systemic artery
(2) More than pCO_2 in tissues
(3) Less than pCO_2 in atmospheric air
(4) Equal to pCO_2 in pulmonary vein

151. Condition when repeated check of blood pressure is higher than 140/90 mm Hg in a person then it is called
(1) Hypertension (2) Heart failure
(3) Hypotension (4) Cardiac arrest

152. The state of heart when it is not pumping blood effectively enough to meet the needs of the body is called
(1) Cardiac arrest (2) Heart failure
(3) Heart attack (4) Angina pectoris

153. How many cellular layer(s) have to be crossed by oxygen to reach the blood present in pulmonary capillaries from alveolar air?
(1) One (2) Two
(3) Three (4) Four

154. In the heart of mammals the bicuspid valve is present between __A__ and __B__.
Choose the **correct** option that fills the blanks A and B respectively.
(1) Left auricle, left ventricle
(2) Right auricle, right ventricle
(3) Right auricle, left auricle
(4) Right ventricle, pulmonary aorta

155. A person with _____ blood group is a universal donor while a person with ____ blood group is a universal recipient.
Fill the respective blanks by selecting the most appropriate option.
(1) O^{+ve}; AB^{-ve} (2) AB^{-ve}; O^{+ve}
(3) O^{-ve}; AB^{+ve} (4) AB^{+ve}; O^{-ve}

156. Human heart is
(1) Myogenic because it is composed of autoexcitable nodal tissues
(2) Neurogenic because ANS can regulate heart beat
(3) Myogenic because it is made of voluntary, cardiac muscles
(4) Neurogenic because it has an extensive nerve supply

157. With the help of a standard ECG we can determine the rate of heart beat of an individual by counting the number of _____ that occur in a given time period.
Select the option to fill the blank **correctly**.
(1) P-waves (2) T-waves
(3) QRS complexes (4) P-Q intervals

CC-115

158. Nearly ___A___ per cent of CO_2 is transported by RBCs whereas ___B___ per cent of it is carried as bicarbonate. Choose the option that **correctly** fills the blanks A and B.

	A	B
(1)	70	20-25
(2)	20-25	70
(3)	70	7
(4)	7	23

(1) (1) (2) (2)
(3) (3) (4) (4)

159. A chemosensitive area is situated adjacent to the rhythm centre which is highly sensitive to
(1) CO_2 only (2) H^+ only
(3) CO_2 and H^+ (4) O_2 and H^+ only

160. In grinding or stone breaking industries, long exposure to dust can cause serious lung damage due to
(1) Proliferation of fibrous tissue
(2) Thickening of muscles
(3) Accumulation of blood in alveoli
(4) Dilation of air pathway

161. Choose the **incorrect** statement among the following w.r.t. ABO blood grouping in human.
(1) ABO grouping is based on the presence or absence of three surface antigens, *i.e.* A, B and O on the RBCs.
(2) The plasma of different individuals contain one or two or both natural antibodies.
(3) During blood transfusion, the blood of donor has to be carefully matched with the blood of the recipient.
(4) Persons with 'AB' blood group are called universal recipients.

162. The primary site for exchange of respiratory gases in humans is
(1) Trachea (2) Bronchus
(3) Primary bronchiole (4) Alveoli

163. The maximum number of action potentials which sino-atrial node can generate in a healthy adult human in normal physiological conditions of the body are
(1) 60-65/minute (2) 70-75/minute
(3) 85-90/minute (4) 100-110/minute

164. Which of the following events does not occur during the joint diastole?
(1) Mitral valves are open
(2) Blood flows from atria to ventricles
(3) Aortic valves are open
(4) Blood enters into right atrium through vena cava

165. An enzyme involved in transportation of CO_2 as bicarbonates and present in high quantity in RBCs, is called
(1) Carbonic anhydrase (2) Carboxypeptidase
(3) Catalase (4) Chitinase

166. The disease which is caused due to the deposition of calcium, fat, cholesterol and fibrous tissue is
(1) Heart failure (2) Atherosclerosis
(3) Cardiac arrest (4) Heart attack

167. In a standard ECG of a normal person, end of T wave represents
(1) Contraction of both the atria
(2) Beginning of atrial systole
(3) Beginning of the ventricular contraction
(4) End of ventricular systole

168. Choose the **correct** option to complete the analogy.
Pulmonary vein : Carries blood towards the heart : : Pulmonary artery : _____.
(1) Carries blood towards right atrium
(2) Carries blood towards left atrium
(3) Carries blood away from the heart
(4) Carries blood towards right ventricle

169. Nodal musculature responsible for initiating and maintaining the normal rhythmic contractile activity of human heart is
(1) Bundle of His (2) AV bundle
(3) Sino-atrial node (4) Atrio-ventricular node

170. Every 100 mL of deoxygenated blood delivers approximately ___ mL of CO_2 to the alveoli under normal physiological condition.
Choose the option which fills the blank **correctly** w.r.t an adult healthy man.
(1) 1 (2) 2
(3) 4 (4) 5

171. Solubility of CO_2 in the blood is ___ times higher than O_2.
Fill the blank with a suitable option.
(1) 120-125 (2) 80-100
(3) 20-25 (4) 40-70

172. Which one of the following formed elements of blood are **not** true cells?
(1) RBCs only
(2) RBCs and platelets
(3) Platelets only
(4) All formed elements are cells

173. Read the following statements and select the incorrect statement about RBCs of adult human
(1) The average life span of an RBC is 120 days
(2) Spleen is known as the 'graveyard of RBCs'
(3) RBCs have a red coloured, iron containing pigment protein called haemoglobin
(4) A healthy individual has 8-10 g of haemoglobin in every 100 ml of blood

174. The partial pressure of carbon dioxide in tissues is
(1) 40 mm Hg (2) 45 mm Hg
(3) 55 mm Hg (4) 75 mm Hg

175. In humans, about 97% of O_2 is transported by
(1) WBCs (2) RBCs
(3) Platelets (4) Plasma in dissolved form

176. Closed circulatory system with 4 chambered heart is found in all of the following, **except**
(1) Birds (2) Mammals
(3) Amphibians (4) Crocodiles

177. The most abundant cells in the human blood in normal conditions are
(1) Erythrocytes (2) Thrombocytes
(3) Leucocytes (4) Platelets

178. Which of the following parts of human brain has cardiovascular center that can moderate the cardiac function?
(1) Mid brain (2) Medulla oblongata
(3) Cerebrum (4) Cerebellum

179. In all of the given cases, there is shift of O_2-dissociation curve towards left except
(1) Low temperature (2) Low H^+ ion concentration
(3) Low pCO_2 (4) Low pO_2

180. A chronic disorder characterized by damaged alveolar wall due to excessive smoking, resulting in reduced respiratory surface is called

(1) Emphysema
(2) Fibrosis
(3) Pneumonia
(4) Common cold

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